

# Lokesh Reddy V

Generative AI Engineer · RAG & Agentic Systems · LLM Evaluation & Guardrails  
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## PROFESSIONAL SUMMARY

Generative AI Engineer with 5+ years of hands-on production experience across regulated financial services, enterprise telecom, and professional services — with the last three years focused on RAG pipelines, agentic orchestration, and LLM evaluation. Track record of cutting regulatory research time by 91%, reducing escalations by 23%, halving infrastructure provisioning cycles, and shipping evaluation frameworks that gate every production release. Treats security, privacy, and auditability as first-class engineering requirements.

## CORE COMPETENCIES

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| LLM Engineering    | GPT-4o, Azure OpenAI, Llama 3, Claude 3.5, LoRA / QLoRA fine-tuning, prompt engineering, structured outputs, tool calling        |
| RAG & Retrieval    | LangChain, LlamaIndex, Pinecone, FAISS, Weaviate, hybrid search, cross-encoder reranking, metadata filtering, source attribution |
| Agentic Systems    | LangGraph, multi-step orchestration, deterministic state machines, controlled tool execution, human-in-the-loop review           |
| Evaluation         | RAGAS, DeepEval, 200+ curated domain test sets, CI/CD quality gates, regression testing, groundedness & relevance scoring        |
| APIs & MLOps       | Python, FastAPI, REST APIs, Docker, Kubernetes (HPA), MLflow, Azure, Databricks, PySpark, Git                                    |
| Security & Privacy | Prompt injection defense, PII detection & redaction, access-aware retrieval, output validation, content controls, audit logging  |

## PROFESSIONAL EXPERIENCE

### Generative AI Engineer

Sep 2026 – Present

Tata Consultancy Services · United States

- Architect and build RAG pipelines over approved regulatory knowledge bases — covering Basel III, CCAR stress testing, DFAST, MiFID II, Dodd-Frank, and internal risk policies — spanning ingestion, semantic chunking, metadata enrichment, access-aware filtering, hybrid retrieval, cross-encoder reranking, grounded generation, and source attribution for 1,800+ risk, compliance, and legal professionals.
- Design multi-step LLM workflows for regulatory document summarization, policy Q&A, change-of-regulation impact analysis, and compliance checking — with explicit state management, structured outputs, validation loops, retries, and mandatory legal-review gates before outputs are distributed to regulated business lines.
- Implement financial-grade security controls: prompt injection defenses, PII and PCI-DSS-aware data redaction, employee-level access scoping via enterprise RBAC, output validation, content controls, and immutable audit trails satisfying OCC examination requirements.
- Build evaluation suites measuring retrieval precision/recall, regulatory answer groundedness, policy adherence, latency (P95), and token cost; integrate threshold-gated quality checks into every CI/CD release pipeline to meet internal model risk governance standards.
- Package AI capabilities as versioned FastAPI microservices deployed within approved Azure Confidential Computing boundaries; instrument with distributed tracing, anomaly alerting, and rollback controls aligned to enterprise model release governance.

### Senior AI Engineer — AI/ML & Agentic Systems

Jan 2024 – Aug 2026

KPMG · United States

- Led development of an agentic audit intelligence platform spanning 12 industry verticals — automating evidence gathering, workpaper linkage, risk flagging, and anomaly detection across client financial statements; reduced average per-engagement manual document review time by 34%.
- Built a production RAG pipeline over audit methodology documentation, PCAOB and GAAS standards, SEC interpretive guidance, and IFRS/ASC pronouncements — enabling engagement teams to surface grounded, cited answers in under 30 seconds vs. 45+ minutes of manual search.

- Developed an agentic due diligence pipeline for Deal Advisory that ingested M&A target financial documents and produced structured risk-finding summaries, financial anomaly flags, and standardized deal-memo sections ready for senior reviewer sign-off.
- Designed LangGraph workflows with mandatory human-review gates before any AI-generated content entered client-facing deliverables — meeting quality-control standards, PCAOB independence rules, and internal review hierarchy requirements.
- Established automated LLM evaluation using RAGAS and DeepEval with 200+ curated accounting-domain test cases and CI/CD quality gates; grew evaluation coverage from 0% to 87% of production flows within six months, blocking 11 regressions before they reached client-facing tooling.
- Fine-tuned Llama 3 (8B) with QLoRA (4-bit NF4) on 12,000 domain samples; improved task F1 from 0.74 to 0.91 and reduced per-query inference cost by 38% through self-hosted vLLM serving — saving ~\$8,400/month at production query volume.
- Implemented engagement-scoped retrieval and confidentiality controls: no cross-engagement document leakage, MNPI and PII redaction before any data entered LLM context, and immutable audit logs meeting data governance and auditor-independence record-retention obligations.
- Deployed Kubernetes HPA autoscaling; system absorbed busy-season spikes at 10x baseline traffic with zero manual intervention and P99 latency under 3.8s.

## Full Stack AI Engineer

Aug 2021 – Dec 2022

BT Group · United Kingdom

- Built a natural language infrastructure provisioning interface for BT's Enterprise Network Engineering division, enabling engineers to generate Terraform and Ansible configurations from plain-English descriptions; cut provisioning cycle time by 60% and manual template errors by 74%.
- Developed a retrieval-backed knowledge assistant over 8,000+ pages of engineering runbooks, network architecture decision records, and incident post-mortems using FAISS and Azure OpenAI; improved first-contact resolution for 100+ engineers by 41%, reducing mean time to find operational guidance from 18 minutes to under 90 seconds.
- Built a NOC intelligence assistant correlating live network alerts with historical incident records and runbooks, surfacing recommended remediation steps; reduced mean time to diagnose recurring fault patterns by 34%.
- Designed multi-step request processing with structured input validation, policy allowlist checks, structured output schemas, and mandatory human approval before any change was executed against production network — zero unauthorized changes.
- Implemented safeguards against unsafe infrastructure instructions: input validation, secrets masking, command allowlists, output sanity checks, and append-only audit logging piped to enterprise SIEM.
- Created 350+ golden retrieval and generation evaluation examples across provisioning, fault-diagnosis, and runbook Q&A tasks; automated pre-production regression tests blocked deployments on failure.

## Application Development Engineer — AI/ML Practice

Nov 2018 – Dec 2019

Tech Mahindra · India

- Developed a BERT-based document classification system for a mortgage lender, achieving 93.4% macro F1 across 12 document categories — replacing a manual rules-based triage process consuming 8+ analyst-hours per 500-document batch.
- Built a RoBERTa-based sentiment analysis pipeline processing 50,000+ daily customer feedback records across email, chat, and web for a retail client; product signals led to three corrective actions within six weeks.
- Designed a transformer-based IT service ticket routing system for 18 support categories; reduced misrouted tickets by 31% and cut mean time to first response by 22%.
- Standardized reusable MLflow experiment-tracking patterns (dataset versioning, parameter logging, model registry, packaging) across 4 concurrent client delivery workstreams.
- Exposed model predictions through Flask REST APIs; onboarded 3 downstream teams within 30 days per deployment.

## EDUCATION

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|--------------------------------------------------------------------------------------------------|---------------------|
| <b>MS, Computer Science</b> · Campbellsville University · Campbell, KY                           | 2023 – 2024         |
| <b>MSc, Information Technology Security</b> · Nottingham Trent University · Nottingham, UK       | Jan 2020 – Aug 2021 |
| <b>BTech, Electronics &amp; Computer Science Engineering</b> · KL University · Vijayawada, India | 2014 – 2018         |